Genetics – Problem Set One

1. In humans, brown eyes are dominant to blue eyes. For each of the crosses below, determine the possible genotypes and phenotypes of the offspring expected. Use a letter of your own choosing.
    a. Homozygous dominant x Heterozygous
    b. Homozygous dominant x Homozygous recessive
    c. Heterozygous x Heterozygous

2. In mice, yellow hair (H) is dominant to normal hair colour (h). Show the expected genotypes and phenotypes that would result from a cross between two heterozygous yellow mice.

3. In peas, round peas (R) are dominant to wrinkled peas(r). If a hybrid round pea is crossed with a wrinkled pea, explain the likely results of the cross.

4. In mice, black coat colour (B) is dominant to brown (b). What are the possible genotypes when a hybrid black mouse is crossed with another hybrid black mouse?

5. In mice, the gene for straight tail is dominant to curly tail.
    a. Find the phenotypes produced when a pure breeding mouse with a straight tail is crossed with a pure breeding mouse with a curly tail. (This is the first generation)
    b. Find the phenotypes produced if two of the first generation are mated.

6. In guinea pigs, rough coat (R) is dominant over smooth coat (r). What are the possible genotypes and phenotypes of a cross between a pure breeding smooth coated guinea pig and a hybrid rough coated guinea pig?

7. In Aberdeen Angus cattle, black coat is dominant to red coat.
    a. If a pure breeding red Aberdeen Angus cow is mated with a hybrid black bull what are the frequencies of coat colour in the offspring produced.
    b. If a farmer produced 60 calves with the above cross, how many calves would be expected to be red?

8. Brown eye colour in humans is dominant to blue eyes.
    a. How often would you expect two brown eyed parents to have a blue eyed child?
    b. If this does occur, what does it tell you about the parents' genotypes?
    c. Can two blue eyed parents ever have a brown eyed child? Support your answer with working.

9. Albino rabbits have white hair and pink eyes. The gene for albinism is recessive. If a pure bred normal rabbit is crossed with an albino one, what

are the different genotypes and phenotypes that could result and their frequencies.

10. Explain how normal rabbits that had an albino parent can both be normal but still produce about one in four albino kittens.

Genetics – Problem Set 2

1. If a pure breeding male black guinea pig is mated with a pure breeding female white guinea pig, what possible offspring could be produced, and in what proportions, given that black coat colour is dominant over white.

2. What phenotypes, and in what proportions, of offspring will be obtained by crossing a hybrid black female mouse with a white male mouse, given that black is dominant to white.

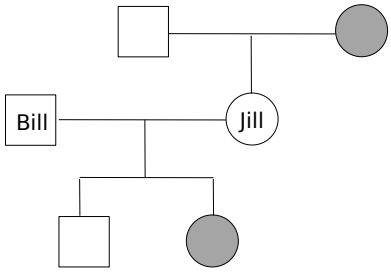
3. For humans, blond hair is recessive to dark hair. What possible hair colour could appear in the children (and with what probability) of a father who is pure breeding for dark hair and a mother who is hybrid for dark hair?

4. In fruit flies, there are 8 chromosomes in a normal body cell. The gene for long wings is dominant over short wings. A hybrid long winged male is crossed with a hybrid long winged female.
    a. How many chromosomes are in the egg of the female?
    b. How many chromosomes are in the muscle cells of the male?
    c. What possible gametes can the female produce in relation to wing length?
    d. What offspring and in what proportions could be produced?

5. In human beings, brown eyes are dominant over blue eyes. Suppose a blue eyed man marries a brown eyed woman whose father was blue eyed. What proportion of their children would you expect to have blue eyes?

6. In *Drosophila melangaster*, the normal wing shape is dominant to the 'dumpy' wing shape.
    a. Two heterozygous flies are crossed. Calculate the expected proportion of both genotype and phenotype.
    b. If the above cross resulted in 240 offspring, how many of each phenotype would be expected?
    c. How many of each genotype would you expect out of the 240 offspring?

7. A brown eyed man (Bill), whose parents were both brown eyed, married a brown eyed woman (Jill) whose father had brown eyes and mother had blue eyes. They have a blue eyed daughter and a brown eyed son. What are all the possible



genotypes of the people mentioned? Use the pedigree on the right to help you.

8. Brown eye colour is dominant to blue. The dominant allele is represented by B. In a family the following has been noted: Bob (blue eyes) and Jane (brown eyes) are brother and sister. Their parents are Bill (brown eyes) and Mary (blue eyes). On the father's side there are uncle John (blue eyes), grandfather Tom (brown eyes) and grandmother Martha (blue eyes). Use a pedigree like the one in question 7 to help you answer the following
    a. What is Jane's genotype?
    b. What is Tom's genotype?
    c. Calculate the probability that the next child from Bill and Mary will have blue eyes.